

Experiment 3-Low Wind: Summary Table

Dropped-Only Quality

| Method   | N(drop) | Precision Mean(m) | Precision P50(m) | Precision P90(m) | R@1m  | R@2m  | R@3m   | Roll  Mean | Roll  P90 | Pitch  Mean | Pitch  P90 | Impact Mean | Impact P90 | Impact P95 |
|----------|---------|-------------------|------------------|------------------|-------|-------|--------|------------|-----------|-------------|------------|-------------|------------|------------|
| PPO      | 2000    | 10.465            | 13.345           | 21.405           | 28.9% | 39.5% | 41.8%  | 4.243      | 10.531    | 2.538       | 4.922      | 0.284       | 0.401      | 0.490      |
| Proposed | 1991    | 1.493             | 0.707            | 1.562            | 72.8% | 93.0% | 94.8%  | 6.603      | 12.960    | 4.640       | 11.087     | 0.298       | 0.466      | 0.560      |
| PPO-GRU  | 1967    | 1.522             | 0.922            | 2.660            | 56.8% | 88.3% | 90.7%  | 8.289      | 15.629    | 3.891       | 8.831      | 0.318       | 0.593      | 0.823      |
| MPC      | 1742    | 2.354             | 2.444            | 2.744            | 0.0%  | 18.8% | 100.0% | 1.600      | 2.504     | 1.177       | 2.231      | 0.134       | 0.135      | 0.140      |

No-Drop Breakdown (done but no release)

| Method   | N(done) | N(drop) | Drop Rate | N(no_drop) | No-drop Rate | Crash Rate | Crash+NoDrop | Crash in NoDrop | Timeout+NoDrop | Both Flags |
|----------|---------|---------|-----------|------------|--------------|------------|--------------|-----------------|----------------|------------|
| PPO      | 2000    | 2000    | 100.0%    | 0          | 0.0%         | 0.0%       | 0            | nan             | 0              | 0          |
| Proposed | 2000    | 1991    | 99.5%     | 9          | 0.5%         | 0.5%       | 9            | 100.0%          | 0              | 0          |
| PPO-GRU  | 2000    | 1967    | 98.3%     | 33         | 1.6%         | 1.6%       | 33           | 100.0%          | 0              | 0          |
| MPC      | 2000    | 1742    | 87.1%     | 258        | 12.9%        | 12.9%      | 258          | 100.0%          | 0              | 0          |

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

| Method   | In MPC-NoDrop | Dropped | Drop Rate | Err>3.0m | LargeErr Rate (in dropped) | Mean Err (in dropped) |
|----------|---------------|---------|-----------|----------|----------------------------|-----------------------|
| PPO      | 258           | 258     | 100.0%    | 151      | 58.5%                      | 10.749                |
| Proposed | 258           | 254     | 98.4%     | 17       | 6.7%                       | 1.702                 |
| PPO-GRU  | 258           | 250     | 96.9%     | 27       | 10.8%                      | 1.501                 |